# Mirror – Chiral Toolkit
User Manual

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## Contents

# 1 Overview

**Mirror** (Chiral Toolkit) is a molecular-structure analysis platform built with Streamlit. It provides six functional tabs for the analysis of porous crystalline frameworks (CIF) and guest molecules (PDB).

## 1.1 Technology Stack

| Component | Technology |
|---|---|
| Frontend | Streamlit (Python 3.12) |
| 3D Visualisation | py3Dmol / stmol / NGL Viewer |
| Crystallography | pymatgen (SpacegroupAnalyzer, Lattice) |
| Numerical | NumPy, SciPy (ndimage, signal) |
| Machine Learning | scikit-learn |
| Plotting | Matplotlib (Agg backend) |
| File parsing | Custom CIF / PDB / DEF parsers |

# 2 Launching the Application

Run from the project root:

```
streamlit run app.py
```

Or double-click `run.bat` on Windows. The app opens at `http://localhost:8501`.

# 3 Tab 1: PDB Enantiomer (Mirror Generator)

Generates mirror images of molecules from PDB or RASPA DEF files.

## 3.1 Workflow

1. Upload a `.pdb` or `.def` file.
2. View molecular analysis: formula, atom count, mass, bounding box, VDW radii, geometric and mass-weighted centroids.
3. Select a mirror plane: **YZ (invert x)**, **XZ (invert y)**, or **XY (invert z)**.
4. The mirrored structure is displayed alongside the original.
5. Download the mirrored PDB or mirrored DEF.

## 3.2 Notes

- Reflection is performed about the geometric centroid.
- For DEF files, only the atom-position block is modified; force-field parameters are preserved.

# 4 Tab 2: CPA Alignment

Canonical Principal Axes (CPA) alignment rotates a molecule so that its principal axes of inertia align with the Cartesian axes.

### 4.1 Workflow

1. Upload a `.pdb` file.
2. Toggle hydrogen visibility and render style (stick, sphere/CPK, line).
3. Select method: **mass-weighted** (inertia tensor) or **geometric** (covariance matrix).
4. Click **Standardize Structure**.
5. Four views are shown: Original, Mirrored (YZ reflection), CPA-aligned, and MCPA-aligned.
6. Each view displays principal moments ($I_1, I_2, I_3$) and the ratio $I_1/I_3$. Degeneracy or near-spherical shape is flagged.
7. Download the CPA or MCPA PDB.

For the mathematical theory, see the separate *Mathematics and Algorithms* document.

The central analysis tab for porous crystals.

### 4.2 Structure Visualization

- Upload a `.cif` file.
- Adjust supercell dimensions ($n_a, n_b, n_c$).
- Toggle cell box and background colour.
- Rendered via NGL Viewer.

### 4.3 Symmetry Analysis

Runs `pymatgen.SpacegroupAnalyzer` across multiple tolerances (`symprec` = 0.01, 0.05, 0.1, 0.2). Reports:

- Best space-group symbol and number.
- Confidence score (fraction of tolerances in agreement).
- Sohncke and enantiomorphic flags.
- Full scan table.

### 4.4 Element-based Channel Map

Detects 1D channels via 2D Voronoi tessellation in the *ab*-plane.

### 4.5 Grid-based Pore Detection & VDW Heatmap

A detailed method producing:

- Distance-field map with pore centre and radius.
- Cylindrical projection of wall atoms (element-coloured).
- Smooth Gaussian VDW heatmap on the $\theta$–$z$ plane.
- FFT-based pore-shape classification: L2, L3, L4, or L6.

### 4.6 Turbo Mode

Ultra-fast alternative using Voronoi tessellation + KDTree. Suitable for real-time interactive use.

## 5 Tab 4: Accessible Orientations

Places a guest molecule inside a host channel and scans all orientations.

### 5.1 Workflow

1. Upload host CIF and guest PDB.
2. Preview four guest orientations: Original, Mirrored, CPA-guest, MCPA-guest.
3. Toggle CPA alignment and select collision method (**voxel** or **vdw**).
4. Extract the centralised channel (periodic replication along $c$).
5. View fit analysis (pore diameter vs. guest box diagonal).
6. Run channel accessibility (Mollweide map, $\theta$–$\phi$ heatmap).
7. Configure scan ($\theta$ and $\phi$ steps, clash scale, mirror plane) and click **Scan Orientations**.

### 5.2 Three-Region Partition

Rolls the guest envelope sphere through the pore cross-section:

- **Green**: envelope fits — all orientations OK.
- **Amber**: only centroid fits — limited orientations.
- **Red**: centroid does not fit.

### 5.3 Pore Explorer

Interactively positions a crosshair and computes L/D orientation spheres. Displays 3D sphere, clearance heatmap, and L vs. D comparison.

## 6 Tab 5: Database Browser

Browse the CoRE-COF 1242 database.

- Search by name or number.
- Table with PLD, LCD, Density, Topology.
- Select a COF to view metadata.
- **Import & Run**: loads the CIF, runs symmetry analysis, and executes grid-based pore detection with full visual output.

## 7 Tab 6: Information

Project goals, planned features, and contact details.

## 8 Batch Scripts

Scripts in `modules/scripts/`:

| Script | Purpose |
|---|---|
| `benchmark_database.py` | Full-database pore detection |
| `batch_pore_morphology.py` | Batch morphology computation |
| `batch_fitness.py` | Host–guest fitness screening |
| `spinol_ml_analysis.py` | ML regression on spinol selectivity |
| `plot_morphology_scatter.py` | CI-coloured scatter plots |
| `plot_topology_pie.py` | Layer-cake topology/SG/sgType |
| `plot_sgtype_ddg.py` | sgType vs $\Delta\Delta G$ |
| `scaling_benchmark.py` | Timing-scaling benchmarks |

## 9 Output Data

Batch results are in `modules/database/benchmark/`:

- `pore_morphology.xlsx`: 1099 COFs × 24 columns (pore radius, CI, CF, curvature, Fourier powers, helicity, etc.)
- `benchmark.xlsx`: 1242 COFs with symmetry and dimensions.
- PNG figures (topology cake plots, SG/sgType distributions, scatter plots, ML diagnostics).